Customer Sales Analysis Report

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Introduction:

This dataset provides a detailed record of online sales transactions, enabling comprehensive analysis for business insights, particularly in customer sales behavior and pattern recognition.

Dataset Description:

Source: The dataset is sourced from Kaggle

(link: https://www.kaggle.com/code/stpeteishii/online-sales-customers-clustering/input)

Rows/Items: The dataset comprises 286,392 rows with 36 variables representing individual sales transactions, including categorical, numerical, time series, geographical, and ordinal data types providing a substantial volume for data-driven decision-making.

Variables:

Categorical: These include status, sku, category, payment_method, Name Prefix, First Name, Middle Initial, Last Name, Gender, full_name, E Mail, Place Name, County, City, State, Region, and Username. Such variables are essential for segmenting the data and analyzing patterns based on qualitative attributes.

Numerical: Variables such as order_id, item_id, qty_ordered, price, value, discount_amount, total, cust_id, age, Zip, and Discount_Percent provide quantitative metrics for analysis. These are crucial for understanding the financial aspects of the transactions and customer demographics.

Special Types: Variables like order_date could be considered as time series data, as they capture the temporal dimension of the transactions, allowing for trend analysis over time.

Dataset Summary:

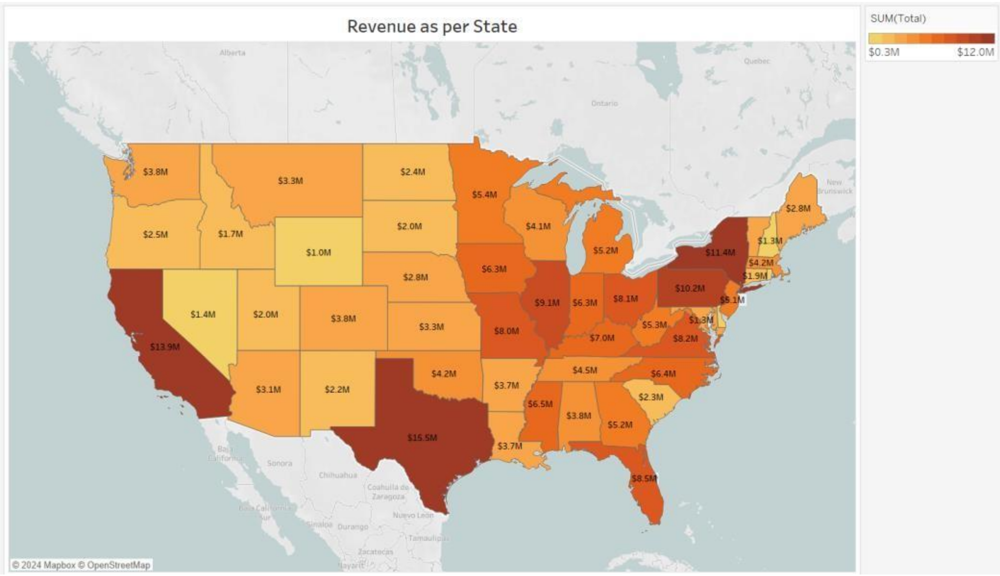
The dataset captures intricate details of each sales transaction, including the transactional date, the number of items ordered, the pricing, discounts applied, and the customer's geographical and demographic data. It provides a foundation for analyzing sales performance, customer behaviour, and potentially, the effectiveness of different sales and marketing strategies. Variables such as qty_ordered and discount_amount can reveal the impact of pricing strategies on sales volume, while geographic data like City, State, and Region offer insights into market penetration and regional sales performance.

Objective:

To optimize marketing strategies and sales efforts by analyzing customer profiles and behaviors in online sales, thereby improving customer satisfaction. The project undertakes a comprehensive analysis of online sales data to uncover insights into customer behaviours, sales trends, and effectiveness of marketing strategies.

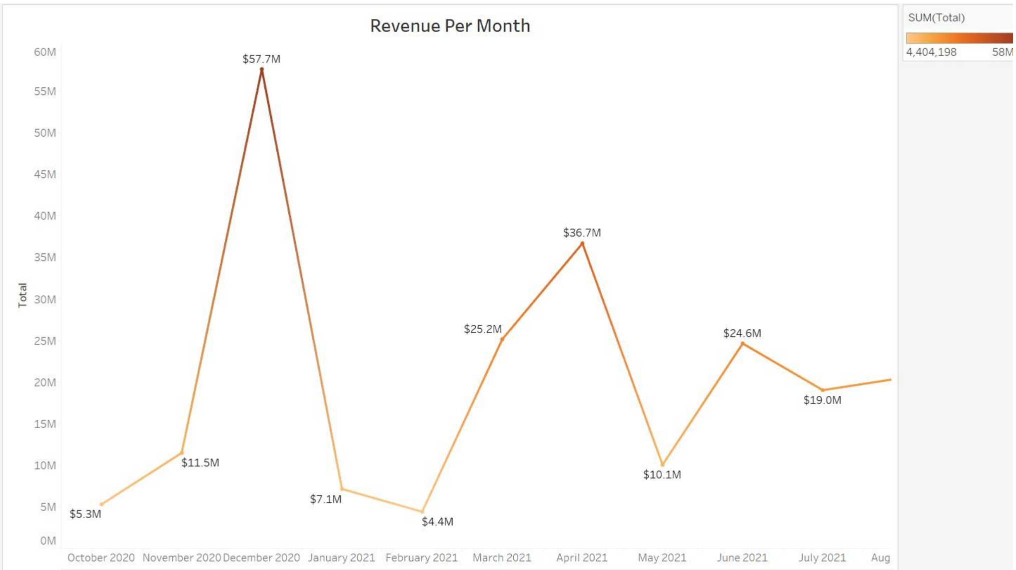
Detailed Visualizations Analysis

1. Revenue as per State (Choropleth Map)



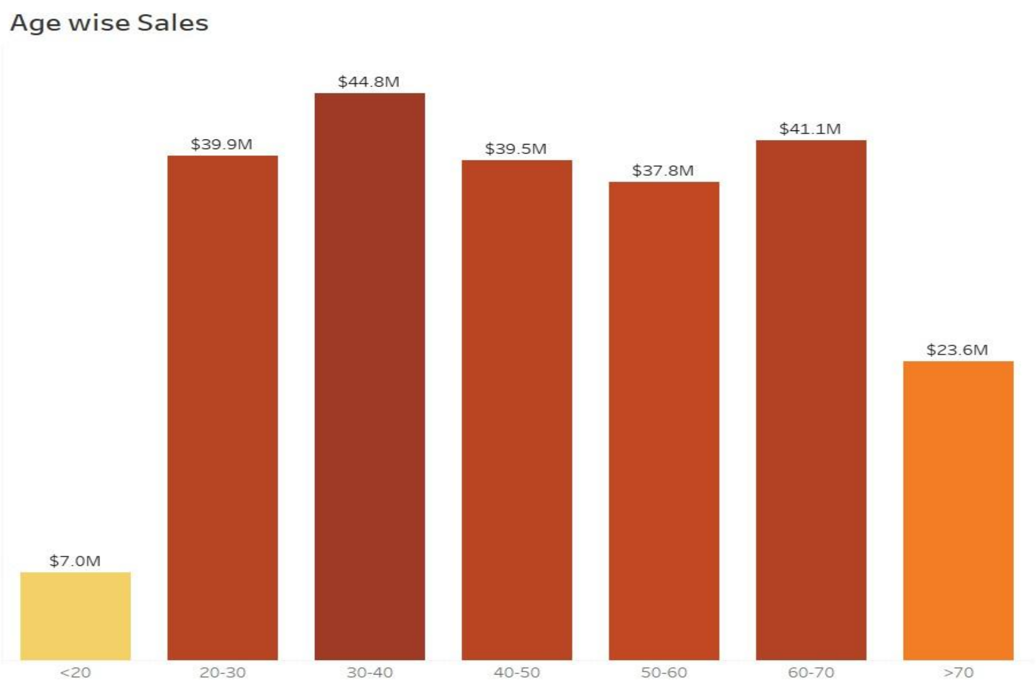
The choropleth map visualizes revenue generation across the United States, highlighting the distribution of sales volume by state. A color gradient from light to dark indicates the range from lower to higher revenue states. Texas and California, with the darkest shades, stand out as the top contributors to sales, showing a substantial market presence and customer spending. In contrast, lighter shades across central and northern states suggest lower sales volumes or market penetration. This spatial representation of data enables businesses to identify key markets and strategize on region-specific marketing and expansion efforts. The map not only serves as a tool for visual assessment but also underlines the geographic disparities in customer buying patterns, which may inform resource allocation for market development.

2. Revenue Per Month (Line Graph)



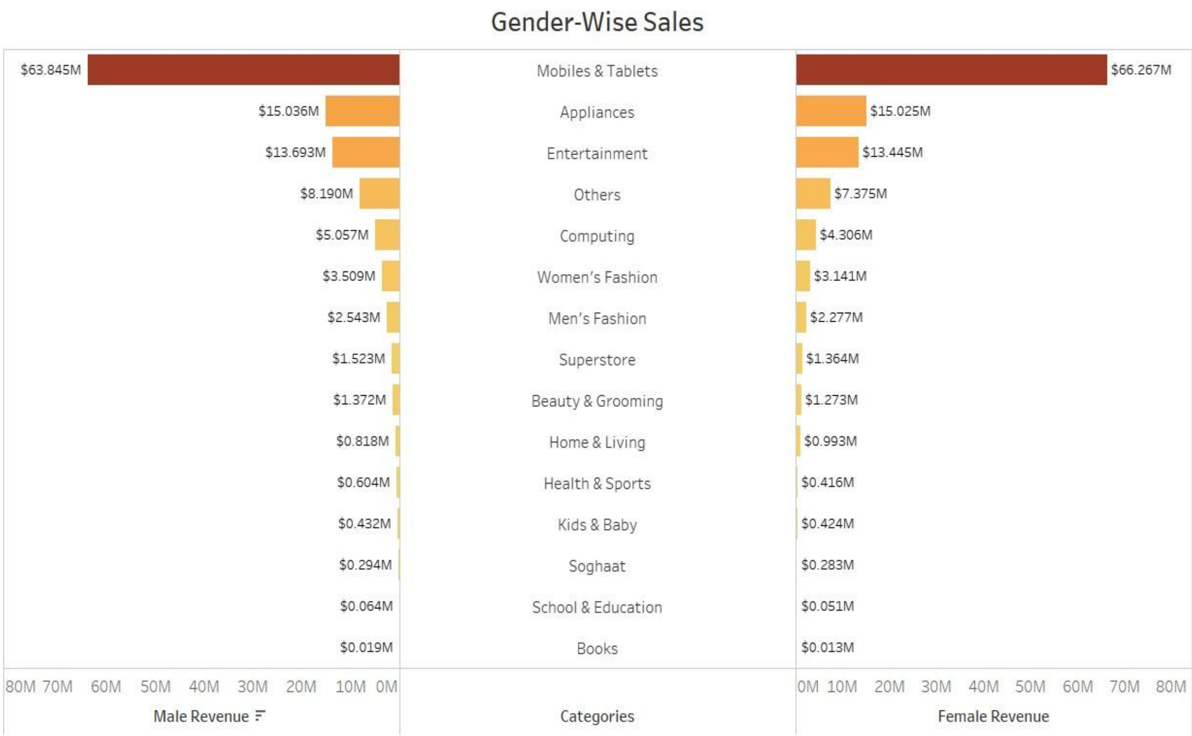
This line graph tracks the changes in revenue over the span of a year, reflecting the dynamic nature of sales across different months. A sharp peak in December indicates a surge in shopping activity, likely due to the holiday season, followed by a stark decrease in the subsequent months. This visualization captures the ebb and flow of consumer spending, showcasing the influence of seasonal factors on purchasing behaviour. The troughs and crests of the line graph can guide businesses in inventory planning and promotional activities, ensuring they capitalize on high-demand periods while managing resources efficiently during slower months. The trend analysis depicted in this graph is pivotal for forecasting and preparing for future sales cycles.

3. Age-wise Sales (Bar Chart)



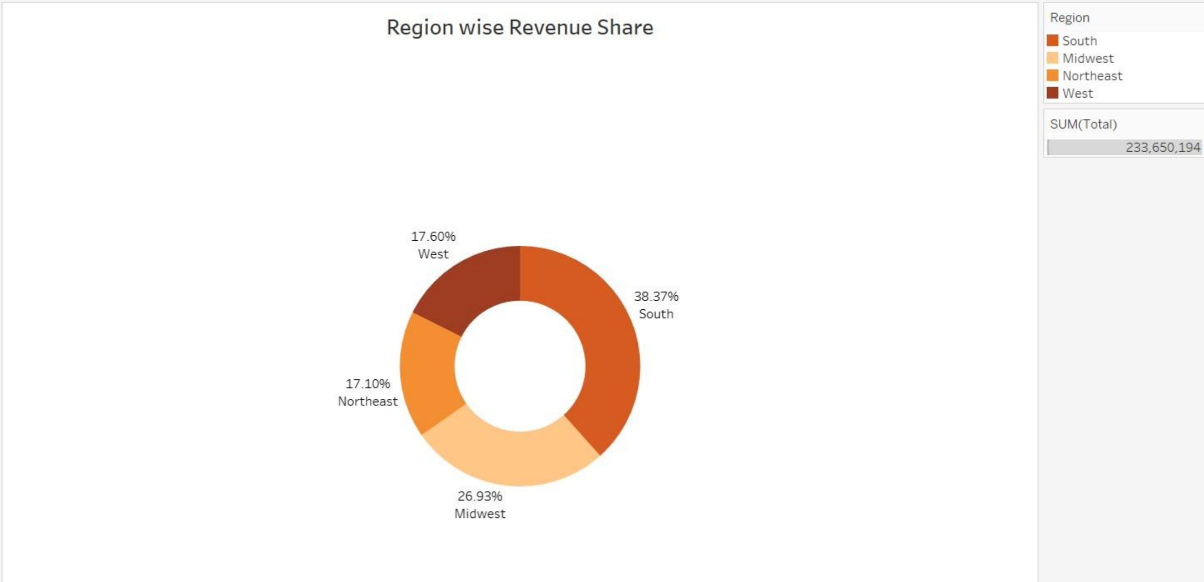
This bar chart categorizes the revenue within various age groups, revealing significant insights into consumer demographics. The highest sales are observed in the 30-40 age group, suggesting this demographic is the most active in online shopping and has a higher spending power. The graph also shows a gradual decline in sales as the age increases, with the lowest sales noted in the under-20 and over-70 categories. This data can assist marketers in targeting their audience more effectively by tailoring campaigns and products to the preferences of the most lucrative age groups. Furthermore, the chart could indicate the potential for growth in markets that are currently underrepresented in sales, such as the youth and senior segments.

4. Gender-Wise Sales (Butterfly Chart)



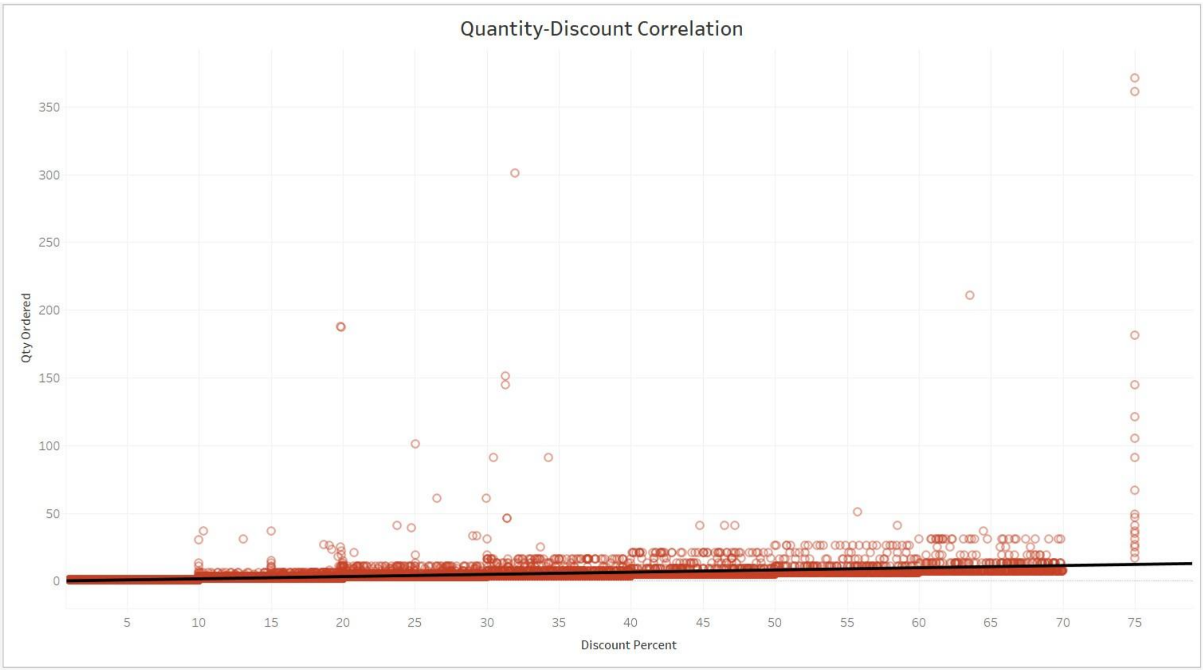
The butterfly chart juxtaposes male and female spending across various product categories, offering a comparative view of gender-based consumption patterns. It highlights that while 'Mobiles & Tablets' is a significant category for both genders, females generally contribute more to sales in almost all other categories. This kind of visualization is instrumental for understanding consumer preferences and can heavily influence product development and marketing strategies. Retailers might use this information to cater to the dominant buying gender in certain categories or to balance the scales by addressing the needs of the less represented gender. The chart not only serves to compare but also to pinpoint disparities and opportunities in gender-specific marketing approaches.

5. Region-wise Revenue Share (Doughnut Chart)

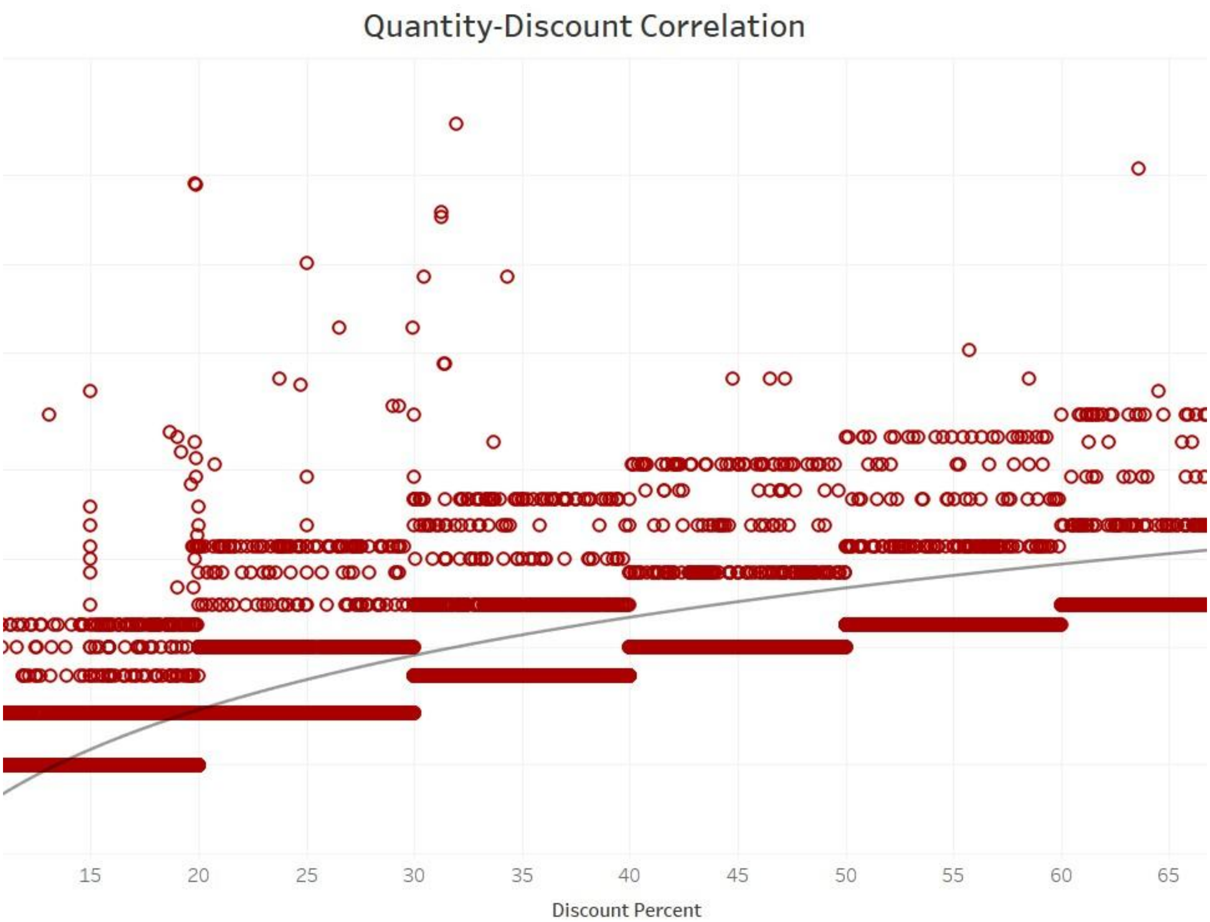


This doughnut chart represents the distribution of total revenue among four key regions. The South holds the largest share, comprising over a third of total sales, which might be attributed to various factors such as population density or economic activity. The Midwest follows, indicating a substantial market presence. The West and Northeast have comparable shares, though less than the Midwest. Such visualizations guide strategic decisions on where to intensify marketing efforts or explore expansion opportunities. They also prompt a deeper look into the socio-economic and cultural dynamics that may influence purchasing patterns in these regions. For instance, regions with higher sales might have more established brand recognition or better distribution networks. This chart is a snapshot of market performance, enabling businesses to evaluate the success of regional strategies. It also aids in forecasting sales and setting benchmarks for growth. Understanding the regional distribution of revenue is vital for aligning sales targets and resource allocation with market potential.

6. Quantity-Discount Correlation (Scatter Plot with trend line)



Scatter Plot with trend line (logarithmic)



The scatter plot illustrates the relationship between the quantity of items ordered and the discount percentage offered. Each dot represents an order, with the position indicating the discount level and the quantity ordered. The trend line suggests a positive correlation; however, the wide spread of dots, especially at lower discount percentages, indicates that the relationship is not strong. This could imply that factors other than discounts are influencing purchasing decisions, such as brand loyalty or product necessity. The visualization aids businesses in understanding how much discounts motivate increased purchases and can help in optimizing pricing strategies. A concentration of dots at lower discount percentages suggests that customers are willing to purchase without significant price reductions. This data is crucial for avoiding unnecessary erosion of profit margins by offering excessive discounts. It also challenges the notion that higher discounts will always result in higher sales volumes. Overall, this scatter plot is an analytical tool for businesses to calibrate their discounting tactics with customer buying behaviour.

7. Heat Map of Order Statuses by Region

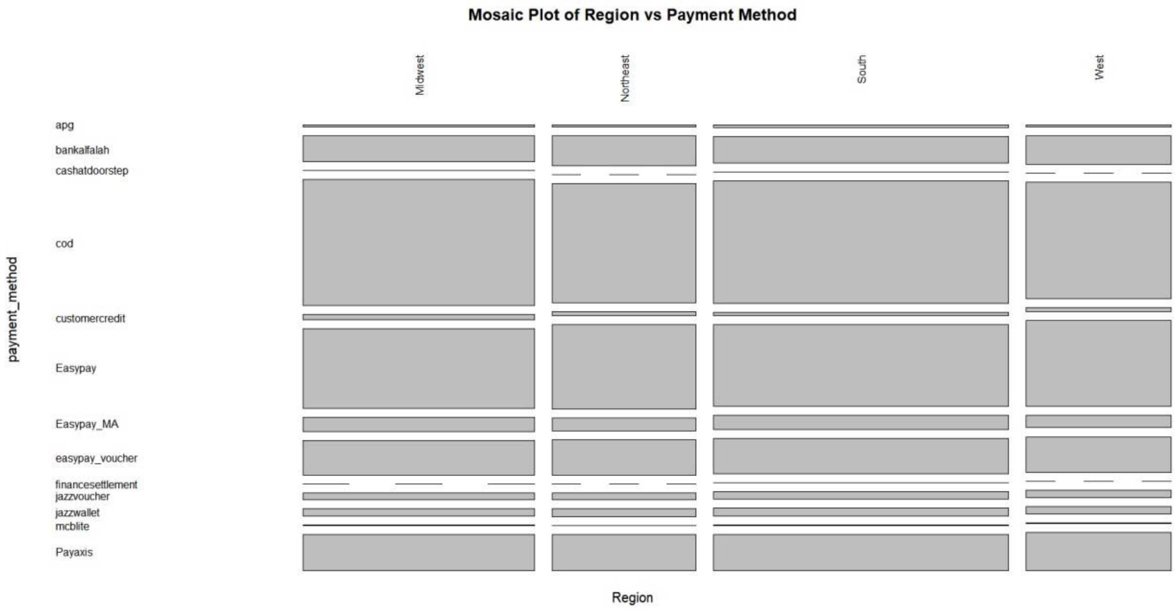
Heat map

Status	Region: Midwest	Northeast	South	West
canceled	94,194	57,288	116,771	59,747
closed	167	56	80	73
cod	1,781	1,444	2,394	924
complete	71,466	44,595	89,615	45,211
holded	10	2	62	16
order_refunded	21,442	10,947	21,609	10,903
paid	1,714	725	1,162	627
payment_review	141	65	60	163
pending	45	9	44	15
pending_paypal		2	6	
processing	7	2	37	22
received	56,987	32,787	66,528	35,882
refund	5,313	2,367	4,773	2,131

The heat map conveys the distribution of order statuses like 'completed', 'canceled', and 'pending' across four regions. Darker colors indicate a higher frequency of a particular status in a region, providing an at-a-glance view of sales operations efficiency. The South exhibits a substantial number of both completed and canceled orders, suggesting active consumer engagement along with areas for improvement in sales processes. This visual can guide businesses in identifying operational strengths and weaknesses in each region, prompting targeted improvements. For instance, a high cancellation rate could lead to reviewing customer satisfaction or product expectations. Understanding the distribution of order statuses also helps in resource allocation for customer service and logistics. This data is crucial for managing inventory, predicting sales trends, and ensuring customer satisfaction.

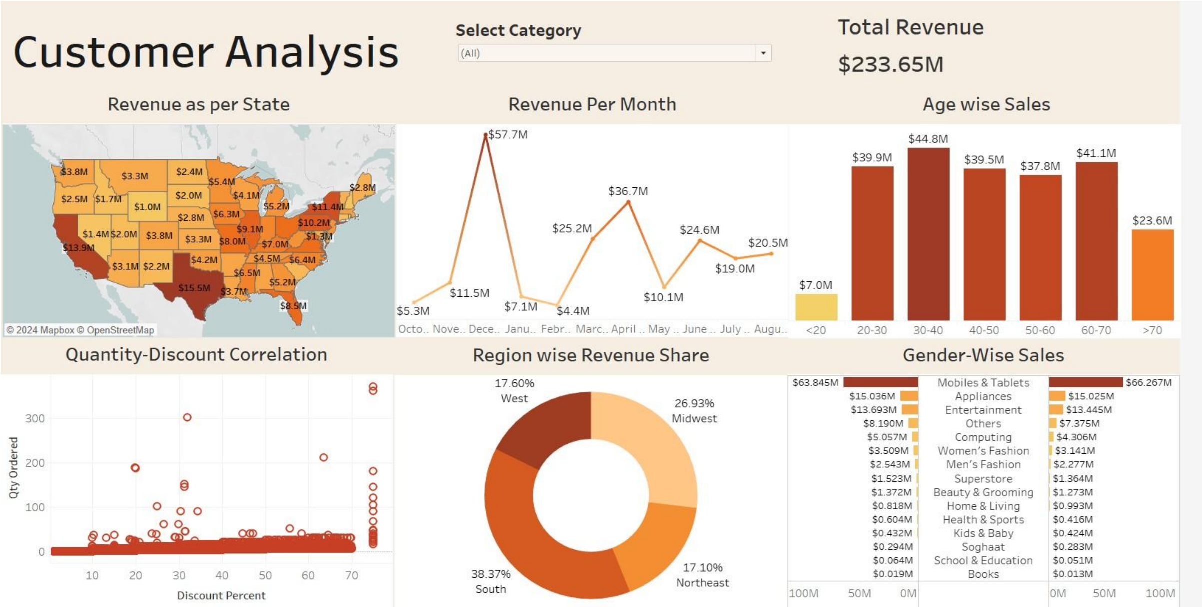
Moreover, it can help in benchmarking performance after implementing new sales or marketing strategies. Overall, the heat map is an essential tool for businesses to maintain high service standards and streamline order fulfillment.

8. Mosaic Plot of Region vs Payment Method



The mosaic plot offers a comparative look at the popularity of various payment methods across different regions. Each tile's size represents the relative frequency of a payment method within a region, with larger tiles indicating more common usage. 'Cash on delivery' (cod) is uniformly popular, implying a trust in this method. Variability in other payment methods like 'bankalfalah' and 'customercredit' may point to regional financial service preferences or availability. This visualization can inform businesses about regional payment preferences, allowing them to tailor their payment options accordingly. It highlights the diversity of consumer behaviour and the necessity for businesses to adapt their payment infrastructure. Such insights are beneficial for forming partnerships with financial institutions and payment platforms popular in certain regions. The plot also illustrates less common payment methods, which might indicate opportunities for promoting alternative payment options. Businesses can use this information to improve the checkout process, reduce transaction friction, and enhance customer experience. Additionally, it may reflect regional economic conditions, internet accessibility, and trust in digital payment methods. By adapting to regional preferences, businesses can potentially increase conversion rates and customer loyalty. In summary, the mosaic plot is a strategic tool for businesses to optimize their payment systems to suit regional trends.

9. Interactive Dashboard Overview



The interactive dashboard consolidates various data visualizations into a single interface, offering users a holistic view of the sales landscape. It allows users to filter and manipulate data to uncover specific trends and correlations that static images cannot provide. The dashboard is instrumental for real-time analysis, where decision-makers can observe the immediate impact of sales activities. Interactive elements such as drop-down menus and clickable charts enable users to explore the data with greater depth and flexibility. The dashboard enhances the user's ability to perform ad-hoc analyses, test hypotheses, and discover insights on-the-fly. This tool encourages a proactive approach to data analysis, fostering continuous improvement in marketing and sales strategies. It also serves as an invaluable communication tool, facilitating data-driven discussions and collaborative decisionmaking. The ease of use invites a broader range of stakeholders to engage with the data, democratizing the analytical process. Ultimately, an interactive dashboard is a key component of a data-centric business culture, driving informed and agile business practices.

Conclusion

The visualizations collectively highlight critical aspects of the sales data, emphasizing the importance of tailoring marketing strategies to customer profiles, regional market nuances, and seasonal trends. This detailed analysis of customer sales through various visualizations offers invaluable insights for business strategy. Understanding the nuanced dynamics of sales across states, months, age groups, genders, and payment methods equips a business with the knowledge to make informed decisions. Tailored marketing approaches, strategic planning, and customer experience optimization are all outcomes of such analysis. The integration of this data into an interactive dashboard further empowers stakeholders, offering a granular view of the sales landscape. These insights are the bedrock for enhancing customer satisfaction, improving sales, and driving business growth.

Appendices

R code:

```
#Problem 1
          essary library utils)

library(   <- read.csv("sales.csv")

# Read the eate a mosaic plot with adjusted label sizes and
sales_data                 orientations
          (~ Region + payment_method, data = sales_data,  main
           = "Mosaic Plot of Region vs Payment Method",  las =
mosaicplot 2,
           cex.axis = 0.8)  # Adjust cex.axis as needed for your specific
plot
```